

✓ *Labsheet 9*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
```

```
uploaded = files.upload()

file_name = list(uploaded.keys())[0]

df = pd.read_csv(file_name, encoding='latin1')

print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Global_Superstore_Sample.csv to Global_Superstore_Sample.csv
Dataset loaded successfully!
Dataset Shape: (12, 20)

DISPLAY DATASET

```
print("\nFirst 5 Records:")
display(df.head())

print("\nDataset Information:")
df.info()

print("\nColumn Names:")
print(df.columns.tolist())
```


First 5 Records:

CHECK MISSING VALUES AND BASIC STATISTICS

Row ID

Order ID

Order Date

Ship Mode

Customer ID

Customer Name

Segment

Country

City

0002

01-12

Class

Sharma

States

Missing Values:

US-

2026-

Second

C-003

Priya

Consumer

United

Los

2

3

2026-

02-03

Class

Singh

States

Angeles

Order ID

0003

02-03

Order Date

0

Ship Mode

US-

0

3

4

2026-

02-18

Standard

C-004

Aman

Home

United

Houston

Customer ID

0004

02-18

Class

Verma

Office

States

Customer Name

0

Segment

0

Country

US-

0

4

5

2026-

03-02

First

C-005

Neha

Consumer

United

Seattle

City

0005

03-02

Class

Gupta

States

V

State

0

Postal Code

0

Region

0

Dataset Information:

Product ID

0

<class 'pandas.core.frame.DataFrame'>

Category

0

RangeIndex: 12 entries, 0 to 11

Sub-Category

0

Data columns (total 20 columns):

Product Name

0

Column

0

Non-Null Count

Dtype

0

Row ID

0

12 non-null

int64

1

Discount

0

12 non-null

object

2

Profit

0

12 non-null

object

3

Order Date

0

12 non-null

object

dtype: int64

3

Ship Mode

12 non-null

object

4

Customer ID

12 non-null

object

5

Customer Name

12 non-null

object

6

Segment

12 non-null

object

Statistical Summary:

Row ID

Postal Code

Sales

Quantity

Discount

Profit

count

12.000000

12.000000

12.000000

12.000000

12.000000

12.000000

9

State

12 non-null

object

mean

6.500000

55053.083333

214.583333

2.416667

0.083333

28.291667

10

Postal Code

12 non-null

int64

std

3.605551

34042.875668

251.110790

1.729862

0.065134

27.358278

12

Category

12 non-null

object

min

0.000000

2128.000000

20.000000

1.000000

0.000000

5.500000

14

Sub-Category

12 non-null

object

25%

3.750000

27480.500000

41.250000

1.000000

0.037500

11.250000

16

Sales

12 non-null

float64

50%

6.500000

67991.000000

125.000000

2.000000

0.100000

19.000000

17

Quantity

12 non-null

int64

75%

9.250000

81401.000000

267.500000

3.250000

0.112500

30.000000

dtypes: float64(3), int64(3), object(14)

max

12.000000

98101.000000

850.000000

6.000000

0.200000

95.000000

memory usage: 2.0+ KB

Column Names:

['Row ID', 'Order ID', 'Order Date', 'Ship Mode', 'Customer ID', 'Customer Name']

KPI CALCULATION

```
total_sales = df['Sales'].sum()
total_profit = df['Profit'].sum()
total_orders = df['Order ID'].nunique()
average_order_value = total_sales / total_orders

print("\n===== KEY PERFORMANCE INDICATORS =====")

print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Orders:", total_orders)
print("Average Order Value:", round(average_order_value, 2))
```

```
===== KEY PERFORMANCE INDICATORS =====
Total Sales: 2575.0
Total Profit: 339.5
Total Orders: 12
Average Order Value: 214.58
```

SALES BY REGION

```
sales_region = df.groupby('Region')['Sales'].sum().sort_values(ascending=True)

print("\nSales by Region:")
display(sales_region)

plt.figure(figsize=(10,5))

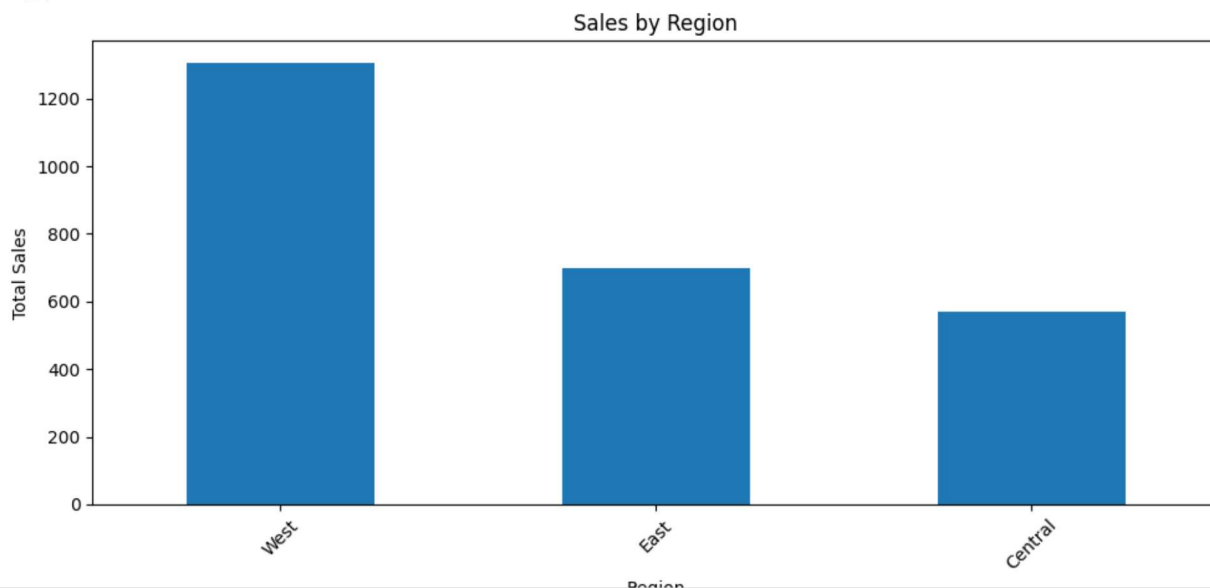
sales_region.plot(kind='bar')

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Sales by Region:

Sales	
Region	
West	1305.0
East	700.0
Central	570.0

dtype: float64



PROFIT BY REGION

```
profit_region = df.groupby('Region')['Profit'].sum().sort_values(ascending=True)

print("\nProfit by Region:")
display(profit_region)

plt.figure(figsize=(10,5))

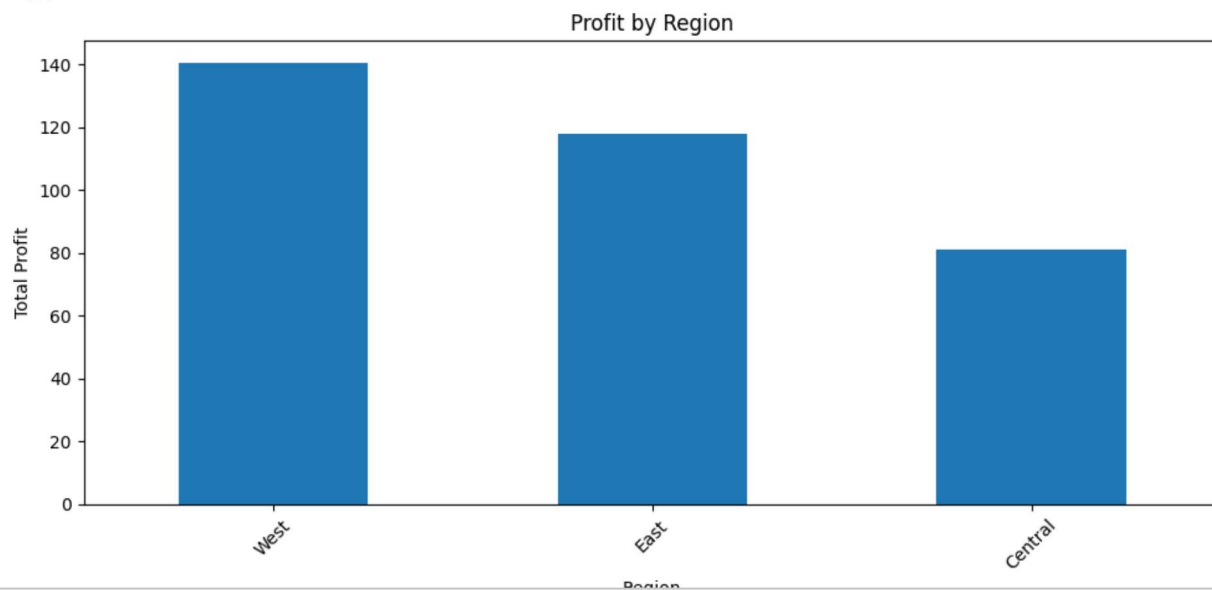
profit_region.plot(kind='bar')

plt.title("Profit by Region")
plt.xlabel("Region")
plt.ylabel("Total Profit")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Profit by Region:

Profit	
Region	
West	140.5
East	118.0
Central	81.0

dtype: float64



SALES BY CATEGORY

```
sales_category = df.groupby('Category')['Sales'].sum().sort_values(ascending=True)

print("\nSales by Category:")
display(sales_category)

plt.figure(figsize=(8,5))

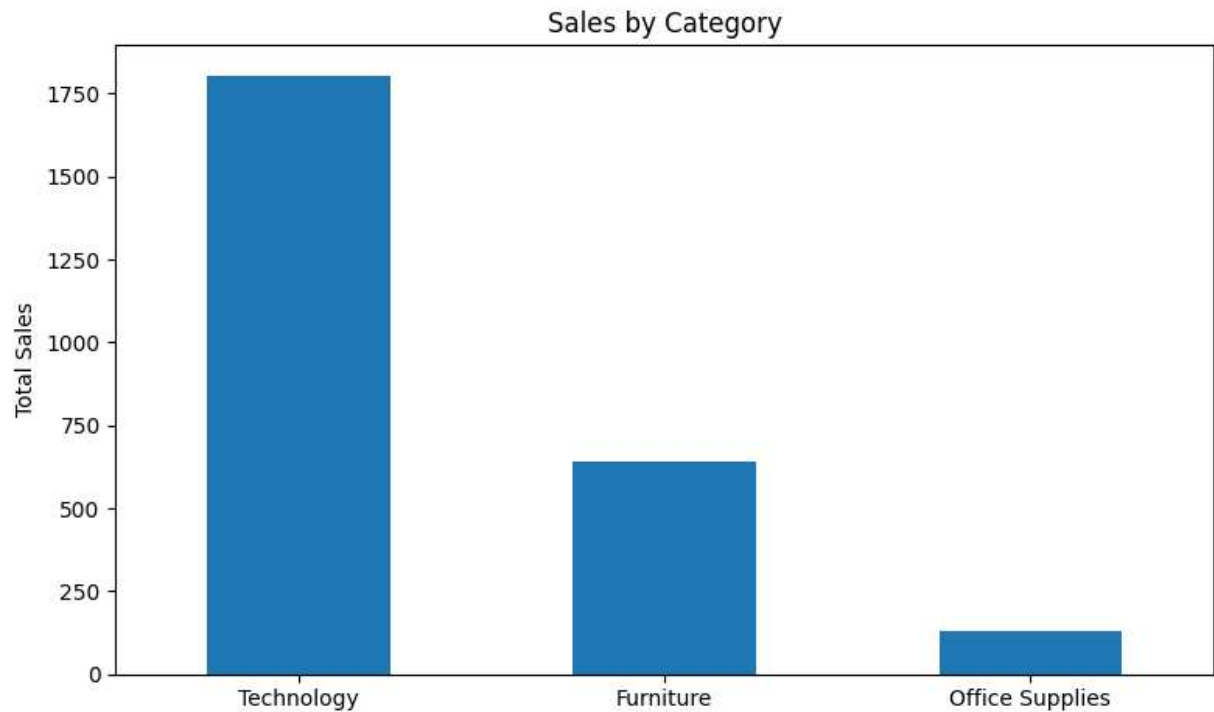
sales_category.plot(kind='bar')

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

Sales by Category:

Category	Sales
Technology	1805.0
Furniture	640.0
Office Supplies	130.0

dtype: float64



PROFIT BY CATEGORY

```
profit_category = df.groupby('Category')['Profit'].sum().sort_values(ascending=
print("\nProfit by Category:")
display(profit_category)

plt.figure(figsize=(8,5))

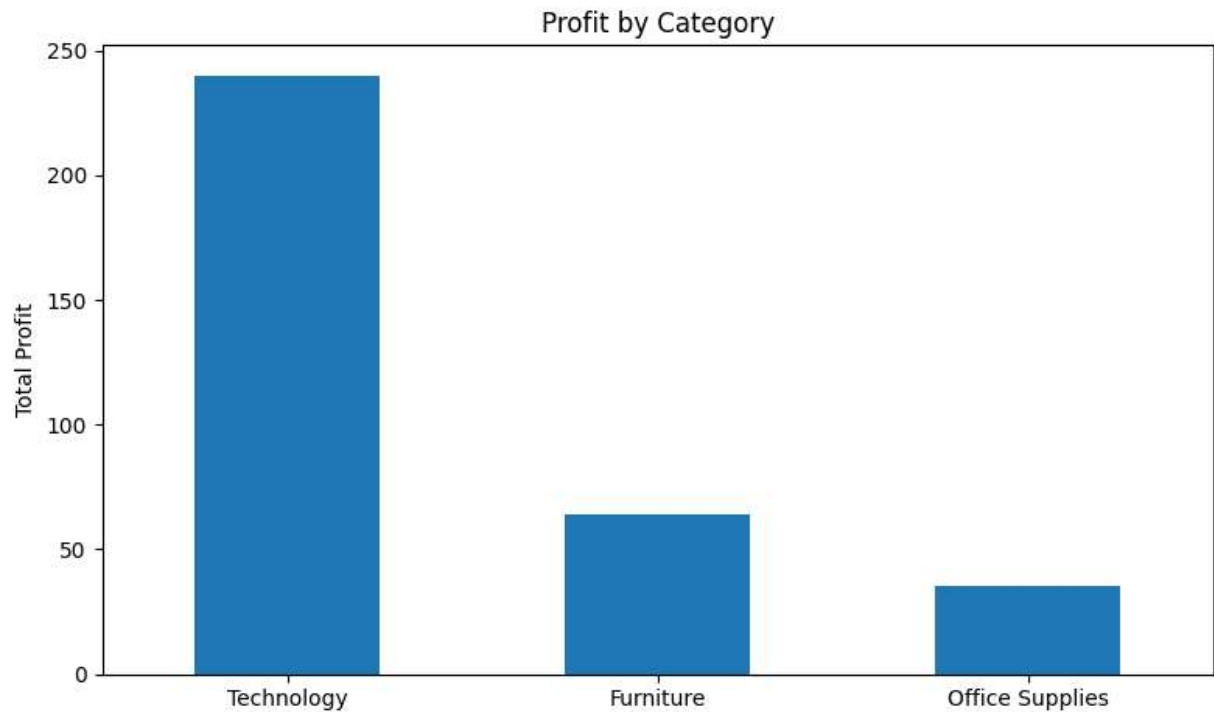
profit_category.plot(kind='bar')

plt.title("Profit by Category")
plt.xlabel("Category")
plt.ylabel("Total Profit")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

Profit by Category:

Category	Profit
Technology	240.0
Furniture	64.0
Office Supplies	35.5

dtype: float64



SALES BY SEGMENT

```
sales_segment = df.groupby('Segment')['Sales'].sum().sort_values(ascending=False)

print("\nSales by Segment:")
display(sales_segment)

plt.figure(figsize=(8,5))

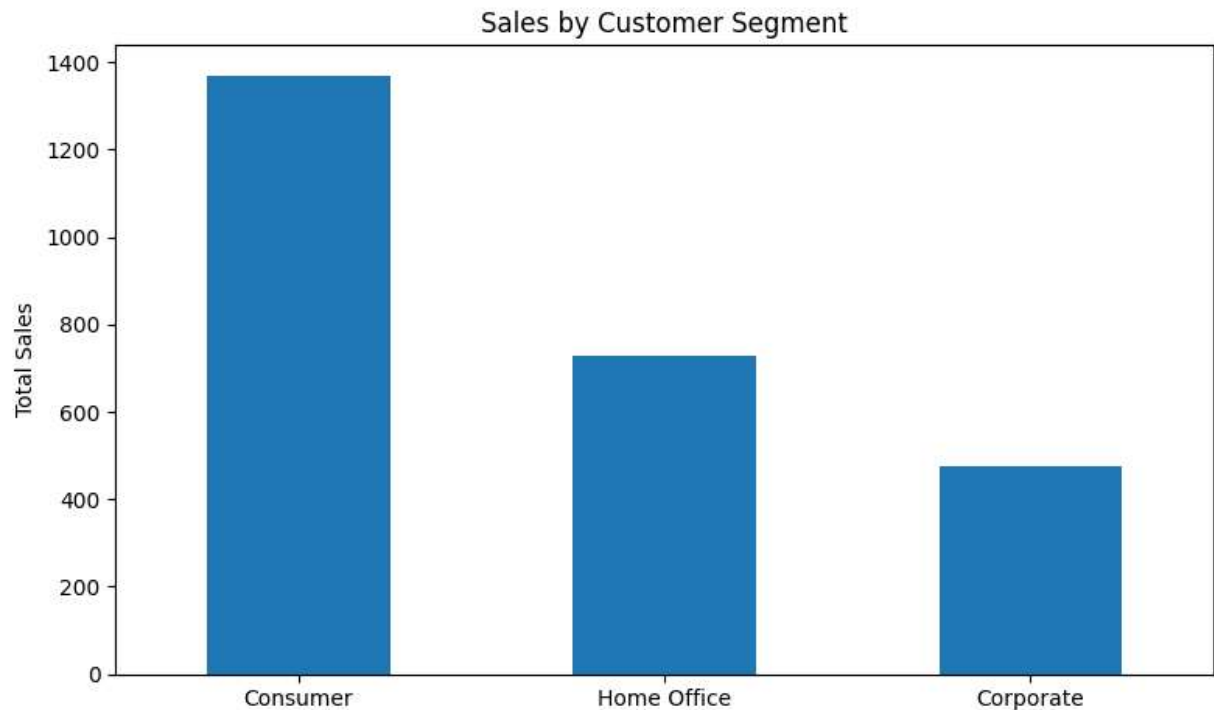
sales_segment.plot(kind='bar')

plt.title("Sales by Customer Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

Sales by Segment:

Sales	
Segment	
Consumer	1370.0
Home Office	730.0
Corporate	475.0

dtype: float64



MONTHLY SALES TREND

```
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')

monthly_sales = df.groupby(
    df['Order Date'].dt.to_period('M')
)['Sales'].sum()

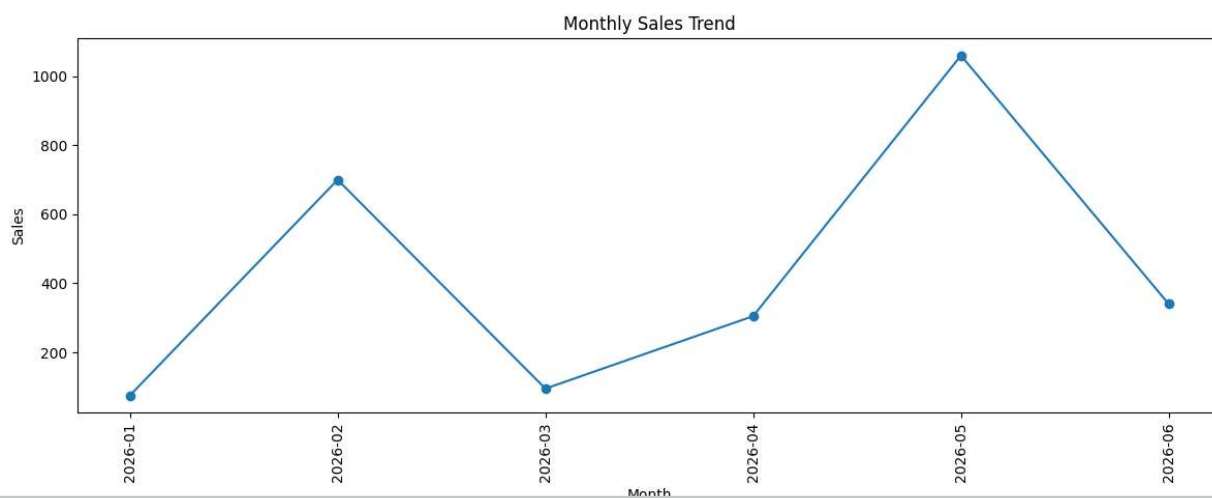
monthly_sales.index = monthly_sales.index.astype(str)

plt.figure(figsize=(12,5))

plt.plot(monthly_sales.index, monthly_sales.values, marker='o')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
```

```
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



```
# 15. TOP 10 PRODUCTS BY SALES
# -----

top_products = df.groupby(
    'Product Name'
)['Sales'].sum().sort_values(ascending=False).head(10)

print("\nTop 10 Products by Sales:")
display(top_products)

plt.figure(figsize=(10,6))

top_products.sort_values().plot(kind='barh')

plt.title("Top 10 Products by Sales")
plt.xlabel("Sales")
plt.ylabel("Product")

plt.tight_layout()
plt.show()

# -----
# 16. TOP 10 PRODUCTS BY PROFIT
# -----

top_profit_products = df.groupby(
    'Product Name'
)['Profit'].sum().sort_values(ascending=False).head(10)

print("\nTop 10 Products by Profit:")
display(top_profit_products)

plt.figure(figsize=(10,6))
```

```
top_profit_products.sort_values().plot(kind='barh')

plt.title("Top 10 Products by Profit")
plt.xlabel("Profit")
plt.ylabel("Product")

plt.tight_layout()
plt.show()

# -----
# 17. CORRELATION HEATMAP
# -----

numeric_columns = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,7))

sns.heatmap(
    numeric_columns.corr(),
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title("Correlation Heatmap")

plt.tight_layout()
plt.show()

# -----
# 18. SALES VS PROFIT
# -----

plt.figure(figsize=(10,6))

plt.scatter(
    df['Sales'],
    df['Profit'],
    alpha=0.5
)

plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

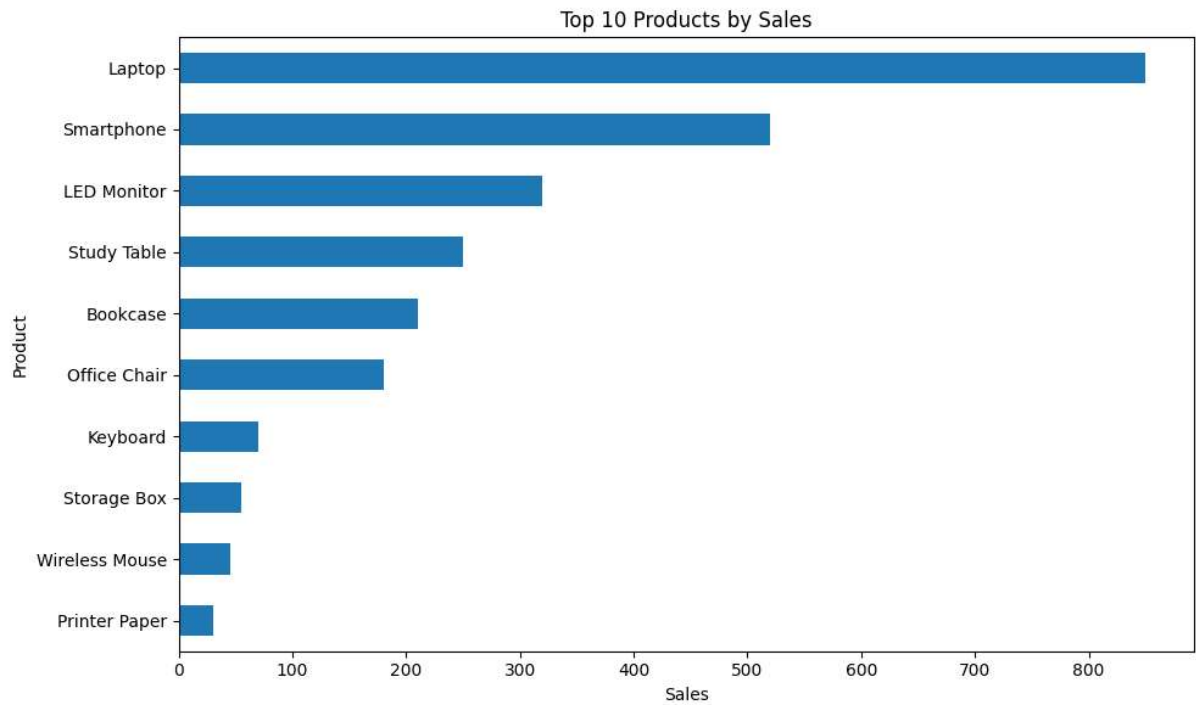
plt.tight_layout()
plt.show()

# -----
# 19. REGION AND CATEGORY ANALYSIS
```

```
# -----  
  
region_category = pd.pivot_table(  
    df,  
    values='Sales',  
    index='Region',  
    columns='Category',  
    aggfunc='sum'  
)  
  
print("\nSales by Region and Category:")  
display(region_category)  
  
plt.figure(figsize=(10,6))  
  
sns.heatmap(  
    region_category,  
    annot=True,  
    fmt='.0f',  
    cmap='YlGnBu'  
)  
  
plt.title("Sales by Region and Category")  
  
plt.tight_layout()  
plt.show()  
  
# -----  
# 20. BUSINESS INSIGHTS  
# -----  
  
highest_sales_region = sales_region.idxmax()  
highest_profit_region = profit_region.idxmax()  
highest_sales_category = sales_category.idxmax()  
highest_profit_category = profit_category.idxmax()  
  
print("\n===== BUSINESS INSIGHTS =====")
```


Smartphone	520.0
LED Monitor	320.0
Study Table	250.0
Bookcase	210.0
Office Chair	180.0
Keyboard	70.0
Storage Box	55.0
Wireless Mouse	45.0
Printer Paper	30.0

dtype: float64

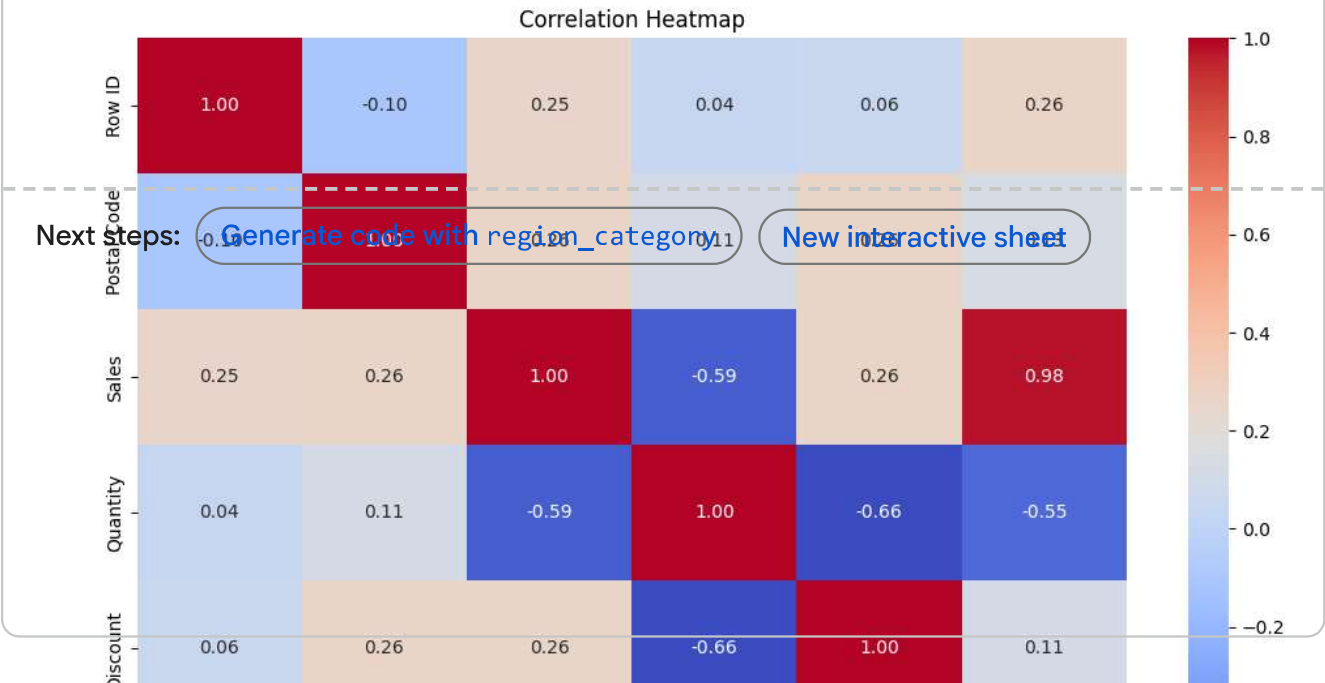
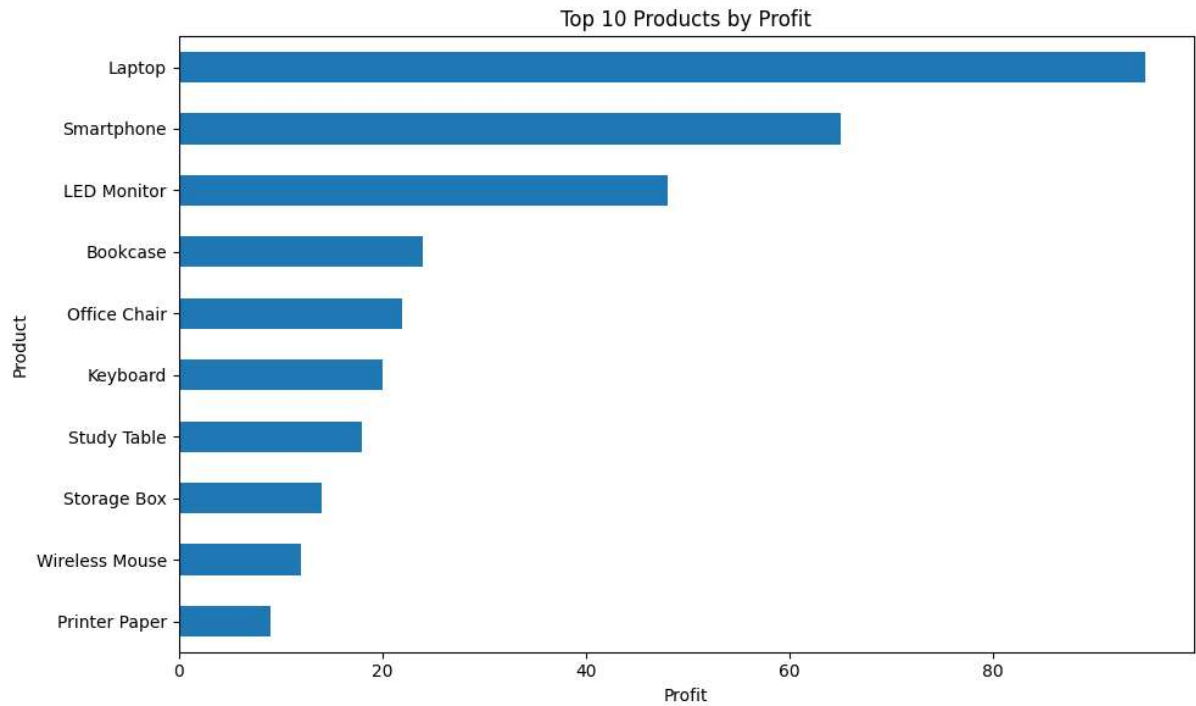


Top 10 Products by Profit:

Profit	
Product Name	
Laptop	95.0
Smartphone	65.0

LED Monitor	48.0
Bookcase	24.0
Office Chair	22.0
Keyboard	20.0
Study Table	18.0
Storage Box	14.0
Wireless Mouse	12.0
Printer Paper	9.0

dtype: float64



1. What is data storytelling?
How is it different from data visualization?

****Answer:****

Data storytelling is the process of combining data analysis, visualizations, and narrative techniques to communicate meaningful insights effectively. It explains not only what the data shows but also why the findings are important and what actions can be taken.

Data visualization mainly focuses on representing data using charts, graphs, maps, and dashboards. Data storytelling goes a step further by adding context, explanation, and a logical narrative to the visualizations.

2. Why is storytelling important in business analytics and decision-making?

****Answer:****

Storytelling is important in business analytics because it helps convert complex analytical findings into simple and understandable information. It enables business stakeholders to understand trends, patterns, problems, and opportunities easily.

Effective storytelling supports better decision-making by explaining the significance of data and suggesting actionable recommendations. It also improves communication between data analysts and business stakeholders.

3. What are the essential

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Quantitative Discount Profit

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components of an effective data story?

****Answer:****

The essential components of an effective data story are:

1. ****Data:**** Reliable and relevant data used for analysis.
2. ****Analysis:**** Examination of trends, patterns, correlations, and anomalies.
3. ****Visualization:**** Charts, graphs, dashboards, and other visual representations.
4. ****Narrative:**** A logical explanation that connects the data and findings.
5. ****Context:**** Background information that helps the audience understand the results.
6. ****Insights:**** Important findings obtained from the analysis.
7. ****Recommendations:**** Actionable suggestions based on the analytical findings.

4. How do Key Performance Indicators (KPIs) enhance business reporting?

****Answer:****

Key Performance Indicators (KPIs) are important measurable values used to evaluate business performance. They help organizations monitor important business metrics and compare actual performance with desired goals.

KPIs make business reports more focused and easier to understand. Examples include total sales, profit, revenue, customer satisfaction, conversion rate, and average order value. By highlighting KPIs, decision-makers can quickly identify performance gaps, opportunities, and areas requiring improvement.

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5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

****Answer:****

Visualizations should be arranged in a logical sequence so that the audience can easily understand the complete data story. A proper sequence helps explain the problem, analysis, findings, and recommendations step by step.

For example, a presentation can begin with an overview of the data, followed by important trends, detailed analysis, key findings, and finally recommendations. This creates a coherent narrative and prevents confusion among the audience.

6. What factors should be considered while selecting visualizations for a business presentation?

****Answer:****

The following factors should be considered while selecting visualizations:

1. ****Type of data:**** Numerical, categorical, geographical, or time-based data.
2. ****Purpose of analysis:**** Whether the aim is to show comparison, trend, relationship, or distribution.
3. ****Audience:**** The visualization should be understandable to the target audience.
4. ****Clarity:**** Charts should be simple and easy to interpret.
5. ****Accuracy:**** The visualization should represent the data correctly.
6. ****Relevance:**** Only important

customer satisfaction, conversion rate, and average order value. By highlighting KPIs, decision-makers can quickly identify performance gaps, opportunities, and areas requiring improvement.

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7. ****Presentation context:**** The visualization should support the overall business story.

7. Explain how dashboards and storytelling complement each other in Business Intelligence.

****Answer:****

Dashboards and storytelling complement each other by combining interactive data exploration with a clear narrative. Dashboards provide multiple visualizations and KPIs in one place, allowing users to monitor business performance and explore data.

Storytelling explains the meaning of the information displayed on the dashboard. It connects different visualizations and highlights important trends, problems, opportunities, and recommendations. Together, dashboards and storytelling help stakeholders understand data and make informed business decisions.

8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

****Answer:****

Several challenges may arise while communicating analytical insights to non-technical stakeholders:

1. Complex technical terminology may be difficult to understand.
2. Large amounts of data can cause information overload.
3. Stakeholders may have different levels of data knowledge.
4. Complex charts may create confusion.

3. **Audience:** The visualization should be understandable to the target audience.

4. **Clarity:** Charts should be simple and easy to interpret.

5. **Accuracy:** The visualization should represent the data correctly.

6. **Relevance:** Only important information should be highlighted.

7. **Presentation context:** The visualization should support the overall business story.

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